

Chapter 17

Cleaning Using Argon/Nitrogen Cryogenic Aerosols

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17.1 INTRODUCTION

Dry (anhydrous) removal of surface contamination has gained increasing importance in semiconductor cleaning applications. Traditional wet cleaning methods employing chemical reaction or dissolution, along with sonic energy or mechanical scrubbing, continue to be effective in removing surface contaminants. Immersion in liquid cleaning agents may also serve to reduce the van der Waals adhesion forces and introduce double layer repulsion forces, thereby promoting the release of insoluble particles from surfaces.¹ However, such methods present increasing disadvantages as circuit dimensions shrink, and as environmental restrictions increase. Among the limitations of wet processing are the high cost and purity requirements of cleaning agents, progressive contamination of re-circulated liquids, re-deposition from contaminated chemicals, special disposal requirements, environmental damage and water consumption, special safety procedures during handling, reduced effectiveness in deeply patterned surfaces

due to surface tension effects (topography sensitivity), dependence of cleaning effectiveness on surface wettability to prevent re-adhesion of contaminants, and possible liquid residue causing adhesion of remaining particles. Wet cleaning agents that depend upon chemical reaction with surface contaminants may also present compatibility problems with new thin film materials, or with more corrosion-prone metals such as copper. In addition, wet cleaning processes typically rely on surface etching to aid in the removal of particles. However, the amount of allowable surface etching is being reduced as devices become smaller and more complex and as new materials are introduced. Table 70a of the 2003 International Technology Roadmap for Semiconductors specifies $<1 \text{ \AA}$ of oxide or silicon loss per cleaning step in 2004 and $<0.5 \text{ \AA}$ loss in 2007.²

Aerosol jet cleaning of surfaces has become a promising technique for the anhydrous removal of particles, or other undesired materials such as films or layers.^{3–7} Aerosol jet cleaning provides gentle, yet efficient, removal of loosely adhering surface contaminants. Carbon dioxide was introduced first as an aerosol jet medium for precision cleaning.^{8–12} This was followed by the introduction of the argon/nitrogen cryogenic aerosol process in 1991, after initial development work at Air Products and Chemicals Inc. and the IBM Corporation.⁷ Aerosol jets use only atmospheric gases for cleaning. Therefore, the process is environmentally sound, and post-clean surface residues are low. Process effluent gases can be vented into the atmosphere. A broad range of applications has been explored, including cleaning silicon wafers, optical mirrors and lenses, magnetic disk drives, and many others. Argon and carbon dioxide can be obtained in bulk quantities and are non-toxic, non-flammable and noncorrosive. Each offers some distinctive advantages and disadvantages. Although the most difficult to produce, cryogenic argon aerosol is perhaps the most suitable for semiconductor applications. Argon solidifies at much lower temperatures than carbon dioxide, but can be produced at extremely high purity. Gaseous argon feed streams can also be filtered to an extremely high level of cleanliness. Also, due to its relatively high vapor pressure, argon can be easily desorbed from surfaces and pumped away under vacuum. As a result, considerable research has been directed toward silicon wafer cleaning using argon aerosols.

Aerosol jet cleaning is inherently a single wafer process. The throughput of single wafer processes is lower than for batch processes such as wet bench cleaning. However, aerosol cleaning can be directly integrated with other single wafer processing tools in a controlled atmosphere or vacuum. Thus, exposure of wafers to atmospheric contaminants such as moisture, and the difficulties created by pump-down and venting of wafer enclosures can be avoided. Contaminants can be immediately removed from the sensitive surface prior to adsorption of moisture or other harmful contaminants, or formation of corrosive reaction products. Also single wafer processing prevents cross-contamination between wafers, as it may occur in batch cleaning processes. The aerosol jet technology has matured considerably and a number of cluster tool cleaning modules, as well as cassette-to-cassette tools, utilizing argon/nitrogen aerosols

have been developed in various laboratories.^a Several of these tools are available commercially. Both front- and back-end applications have been explored and several applications have been implemented in full production at several IC fabrication facilities.^b The argon aerosol jet has now emerged as a complementary technology to liquid chemical cleaning, and several applications have been identified in the silicon wafer processing industry.

This chapter describes the methods used to produce argon/nitrogen cryogenic aerosol jets, approximate operating costs, and some examples of the effectiveness of these aerosols in removing silicon wafer surface contamination. However, the technique is applicable in removing contaminants in other areas. Although operating conditions for a cleaning tool are presented, these conditions are merely illustrative, and should serve only as an approximate guide in generating cryogenic aerosols for cleaning.

17.2 AEROSOL JET CLEANING MECHANISMS

The physical mechanism by which aerosol jets remove submicrometer surface contamination is not fully understood. Theoretical work has been carried out in an attempt to characterize the formation of the aerosol cluster size distribution and velocity distribution.¹³ Solid projectile particles are formed from a cleaning medium and are directed at high velocity toward the contaminated surface, as illustrated in Figure 17.1. The jet is produced by expanding a pressurized gas, liquid, or supercritical fluid through a nozzle. The fluid may consist of a single component or a mixture of substances. Depending on the initial phase and details of the nozzle design, various processes, such as nucleation and growth, vapor flashing, or liquid atomization may occur. The size distribution, number density and morphology of the frozen aerosol particles vary greatly with the particle formation mechanism and thermodynamic conditions (pressure, temperature, flow rate, etc.). The result is a subsonic or supersonic aerosol jet. Cleaning is accomplished through a process of colliding the aerosol particles at high velocity against the surface, as shown in Figure 17.1. The aerosol particles

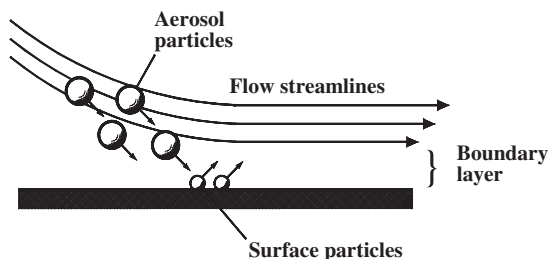


FIGURE 17.1 Particle motion during aerosol jet cleaning.

strike contaminant particles and films on the surface. The removal efficiency of the contaminants is closely related to the aerosol particle size and velocity, the jet temperature, and other parameters.

Possible mechanisms by which aerosol jets effect surface cleaning include the following:

- *Energy transfer to contaminants*—As in conventional sandblasting, collisions between the aerosol particles and the surface cause a direct transfer of energy to the contaminant.⁷ If the transferred energy is sufficient to overcome the contaminant's adhesion energy, it may be released from the surface.
- *Vapor burst*—Cryogenic aerosol particles may rapidly evaporate upon striking a warmer surface.¹⁴ The resulting vapor burst may produce sufficient hydrodynamic shear forces to release contaminants.
- *Reduced adhesion forces*—Particle dislodgment may also be enhanced by reduced van der Waals forces at low temperatures.¹⁵
- *Thermophoresis*—The gas phase of the aerosol impinges against the surface and flows across it, forming a thin boundary layer. If the surface temperature is higher than that of the gas, the released contaminant may be repelled from the surface by thermophoresis.¹⁴ The gas phase of the aerosol carries away the released contaminant.
- *Hydrodynamic shear*—The gas may also provide sufficient hydrodynamic shear force to dislodge larger contaminants.¹⁶ However, the dimensions of the surface contaminant may be so small that it exists completely within the low velocity boundary layer, as shown in Figure 17.1. In this case the gas phase alone cannot remove the microcontaminants because of insufficient shear force. However, the aerosol particles have sufficient inertia to cross the surface boundary layer and strike the contaminant.
- *Thermal contraction*—Since cryogenic aerosols induce cooling of the surface, differential thermal contraction between thick contaminant films and the substrate may crack the films into loosely adhering debris. This debris may then be removed by a combination of particle impact and hydrodynamic shear.^{7,17}

17.2.1 Adhesion and Hydrodynamic Forces

Contaminant particles adhere to surfaces through a combination of the van der Waals, electrical double layer, electrostatic image, capillary, and gravitational forces.¹⁸ For microscopic particles, in the absence of covalent bonding or capillary forces, the van der Waals force typically predominates. For a spherical particle near a flat surface the magnitude of the van der Waals force F_{ad} varies approximately with the particle diameter, $F_{ad} \propto d$:

$$F_{\text{ad}} = Ad/12z^2 \quad (17.1)$$

where A is the Hamaker constant and z is the particle-surface separation distance, generally assumed to be 4 Å. Gas or liquid jets are sometimes used to remove particles from surfaces. When a surface particle is exposed to a fluid flow it experiences a hydrodynamic shear force F tending to set the particle in motion. For a spherical particle in a shear field in the vicinity of a surface the hydrodynamic force is proportional to d^2 . Since $F/F_{\text{ad}} \propto d$, the ratio of removal to adhesion forces decreases with decreasing particle size. Therefore, the ability of fluid flow alone to remove surface particles diminishes as particle size shrinks.

Many damaging surface particles can have dimensions considerably less than the boundary layer thickness, as illustrated in Figure 17.2. Such particles are exposed to a lower average flow velocity U than larger particles. The hydrodynamic shear force on a particle decreases in direct proportion to the average flow velocity, $F \propto U$. Therefore, the hydrodynamic force on such small particles is further limited. This further reduces the ability of a fluid flow to overcome surface adhesion forces in the case of small particles. It is, therefore, not surprising that submicrometer particles are difficult to remove using hydrodynamic cleaning processes such as pressurized gas and liquid jets, or brush scrubbing. Such difficulties in particle removal can be overcome by transferring energy directly to the adhering particle through aerosol impingement.

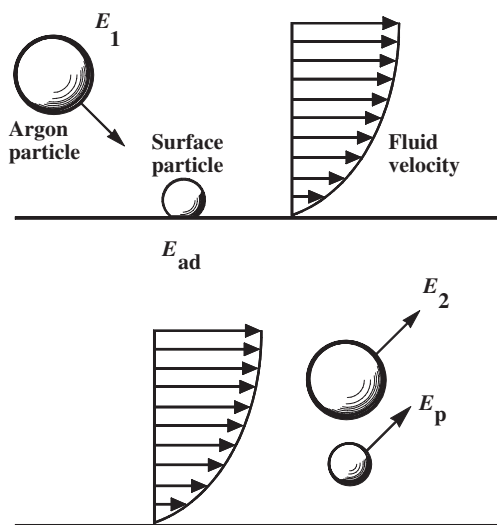


FIGURE 17.2 Particle motion and fluid velocity during aerosol jet cleaning. E_1 is the initial kinetic energy of aerosol particle, E_2 is the final kinetic energy of aerosol particle, E_{ad} is the adhesion energy of surface particle, E_p is the kinetic energy of released surface particle.

17.2.2 Particle Collision

The energy required to separate a particle from a surface is known as the adhesion energy, E_{ad} . This energy is a function of the surface and interfacial energies of the two objects, and must be overcome in order for a contaminant particle to be removed. The magnitude of this energy tends to decrease with particle diameter¹⁹:

$$|E_{ad}| \propto d \quad (17.2)$$

It is believed that aerosol jet cleaners release contaminants from surfaces through a process of direct energy transfer to the adhering particles. This transfer of energy occurs during collisions between aerosol particles and surface particles. An argon aerosol particle approaches the surface just before the collision with an initial kinetic energy E_1 as shown in Figure 17.2. The potential energy of the incoming and rebounding particles can be comparatively negligible compared to their kinetic energy.²⁰ During the collision the aerosol particle loses energy through internal plastic deformation and through energy transfer to the contaminant. The rebounding aerosol particle moves away from the surface with kinetic energy E_2 , and the contaminant particle moves away from the surface with kinetic energy E_p . Particle dislodgment from the surface requires that the energy transferred during the interaction be sufficient to overcome the contaminant's adhesion energy. Neglecting deformation losses, the kinetic energy of the incoming argon particle must exceed the adhesion energy as follows:

$$E_1 > |E_{ad}| + E_2 \quad (17.3)$$

When the above condition is met, the energy threshold for removing the surface particle is achieved. Simple conservation of energy can be applied to an aerosol projectile having mass M . In this case the minimum, or threshold approach velocity needed to re-suspend a surface particle is given by:

$$V_{\text{threshold}} = [(2|E_{ad}|/M) + V_2^2]^{1/2} \quad (17.4)$$

where V_2 is the velocity of the rebounding aerosol particle. This threshold for particle removal is sometimes expressed in terms of a minimum aerosol Stokes number, $Stk_{\text{threshold}}$.²¹

The kinetic energy E_1 available for particle dislodgement in an argon aerosol can be estimated. The mean jet velocity of a typical argon aerosol is estimated to be <100 m/s and experimental tests have shown that the size of nucleated argon particles is of the order of $1 \mu\text{m}$. The size of the argon particles was estimated by exposing a soft film of unbaked photoresist to the aerosol. The size of the resulting crater-like surface imperfections was then measured under SEM. From this measurement, the approximate size of the argon particles was determined. Under these conditions, the kinetic energy available for dislodgment by a single aerosol particle is no more than about 4×10^{-12} J. However, it can be shown that this kinetic energy can exceed the adhesion energies of typical contaminant

particles.²² And as will be seen below, it has been verified experimentally that this kinetic energy is sufficient to overcome the adhesion of some process-generated semiconductor contaminants.

Other experimental studies have attempted to measure the aerosol size and velocity directly using laser diffraction techniques.²³ These studies measured mean aerosol cluster velocities of 24–61 m/s, with peak velocities up to 100 m/s.

Smaller, submicrometer surface contaminants tend to have lower adhesion energies than larger contaminants [see Eq. (17.2)]. Therefore, when direct energy transfer is used to remove contaminants, smaller objects tend to be removed more effectively than larger objects. Experimental tests have, in fact, shown a tendency toward more efficient removal of smaller surface particles than larger particles by argon aerosol jets. This size dependency is the opposite of that encountered with force-dependent cleaning processes, such as single phase jets, which are limited in the smallest size particles they can remove. A further improvement in the removal efficiency of smaller particles has been demonstrated at lower cleaning chamber pressures. This effect is generally attributed to the production of a smaller size distribution and higher velocity distribution in the aerosol.

Although other studies²⁴ have also shown a reduction in the rate of removal with increasing particle size, this effect can be explained through kinetic considerations (Section 17.5.1). Semiconductor applications requiring efficient removal of submicrometer surface contaminants, have, therefore benefited from aerosol jet cleaning.

17.3 PRODUCTION OF ARGON/NITROGEN CRYOGENIC AEROSOL JETS

Figure 17.3 shows a schematic flow diagram of the argon aerosol jet process. A pressurized mixture of high-purity gaseous argon and nitrogen is first filtered and then pre-cooled in a cryogenic heat exchanger. The lower boiling point nitrogen remains in the gas phase and acts as a carrier medium to impart high velocities to the argon particles. The nitrogen also serves as a diluent to control the size of the argon aerosol particles, and permits high expansion ratios which enhance the Joule–Thomson cooling effect, while reducing argon consumption. In applications requiring precision cleaning, such as semiconductor substrate surfaces, extremely pure feed gases and extremely clean components must be used. Trace molecular impurities such as various organic compounds can condense into aerosol particles at cryogenic temperatures. The cryogenic cooler can serve as a cold trap to remove condensed impurities. However, some contaminants may pass through the cooler and be deposited as new surface particles on the sensitive substrate. Argon and nitrogen can be obtained as high purity gases having <1 part per billion molecular impurities. Such high purity feed gases can be supplied from high-pressure cylinders or from bulk gas systems at pressures >7 bar.

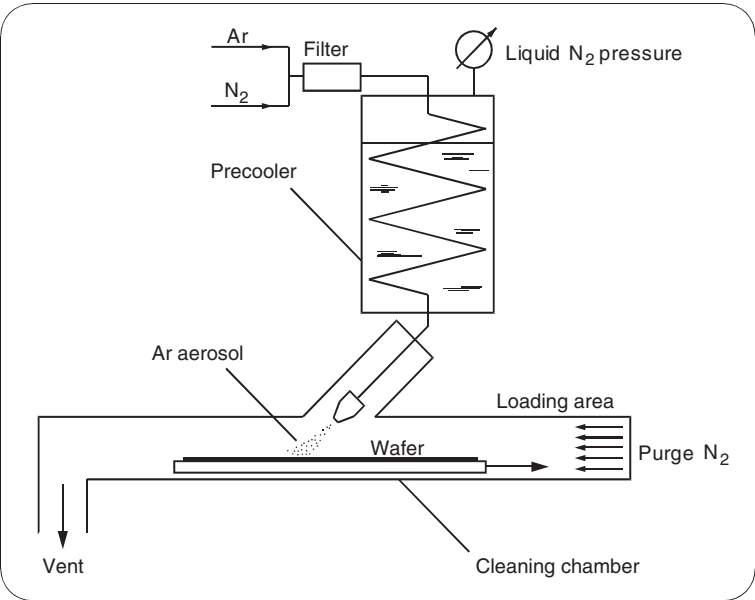


FIGURE 17.3 Schematic flow diagram of aerosol jet cleaning system.

The pre-cooling step can be illustrated schematically using the argon phase diagram in Figure 17.4. Although this diagram shows the phase behavior of pure argon, it can be used to approximately illustrate the aerosol formation process when nitrogen is present in a small proportion. The argon/nitrogen mixture may be pre-cooled to a temperature slightly above the argon dew point in the cryogenic cooler. For the case of a gas mixture containing 90% argon and 10% nitrogen at 5 bar total pressure, the argon dew point is ~105 K. However, the heat exchanger temperature may also be set at the argon dew point to partially

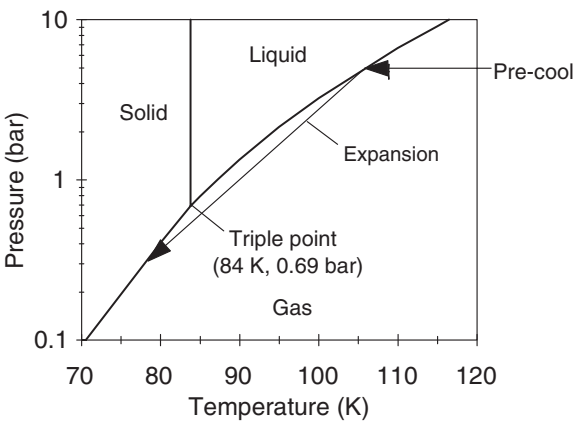


FIGURE 17.4 Argon phase diagram.

liquefy the argon before the expansion step. Best results are obtained if a gas/liquid mixture is delivered to the nozzle.

The pre-cooled mixture is then expanded in a free jet to a lower pressure. The associated Joule-Thomson cooling condenses and solidifies microscopic argon particles in the free jet. The solid argon particles nucleate directly from the gas phase if the expansion passes below the argon triple point (0.69 bar, 84 K) as shown in Figure 17.4. If the expansion passes above the triple point, then the argon first condenses into liquid droplets before freezing into solid argon ice.

If argon liquefaction first occurs in the cryogenic cooler, then argon droplets may be formed through subsequent atomization in the nozzle. These droplets may then freeze at the lower temperatures of the expanded free jet before striking the surface. Experimental studies²³ have shown that hydrodynamic breakup and effervescent flashing followed by evaporative cooling of the liquid component of the stream is a major source of the solid aerosol clusters. Photographs showing the expansion and breakup of the liquid component near the nozzle are shown in Figure 17.5. Solid aerosol particles are preferred over liquid droplets for cryogenic surface cleaning. Solid (ice) particles suffer less deformation during collisions than droplets, and should, therefore, provide a more efficient transfer of energy to surface contaminants.

17.3.1 Equipment Requirements

The high-purity argon and nitrogen feed gas streams are first blended in a manifold to produce a specific mixture ratio. Mass flow controllers are typically used to create a selected argon/nitrogen blend. Microporous membrane

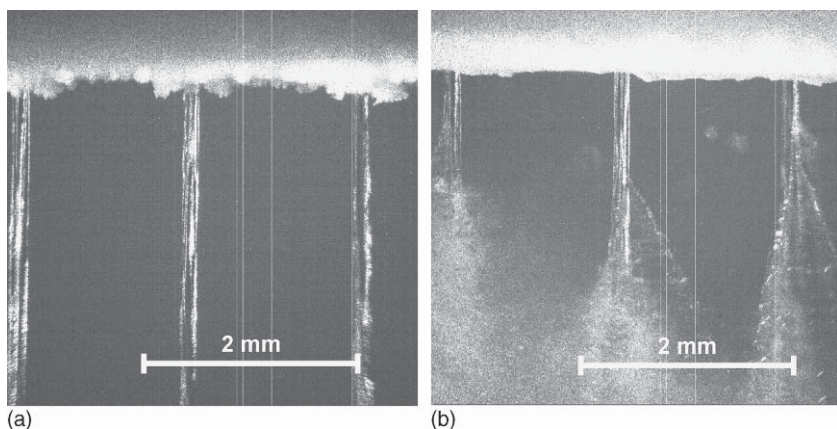


FIGURE 17.5 Photographs of cryogenic aerosol jet formation as the liquid/gas mixture emerges from the nozzle holes into a chamber pressure of (a) 300 torr and (b) 70 torr. At higher chamber pressures the liquid stream breaks up by flow instability, while at lower chamber pressures the liquid stream appears to break up by effervescent flashing.

filtration of the mixture is then performed prior to the pre-cooling step. Such filters can provide an extremely low level of particulate contamination in process feed gases. The cryogenic cooler may consist of a pressurized liquid nitrogen reservoir containing a submerged feed gas tube as shown in [Figure 17.3](#). Boiling nitrogen provides a controllable constant-temperature cooling medium. The temperature in the heat exchanger is set by controlling the pressure of the boiling liquid nitrogen (typically 2.40–4.7 bar). The boiled-off liquid nitrogen must be replenished during operation of the cleaner. An alternative cooler design uses closed-cycle cryogenic refrigeration rather than liquid nitrogen. In this system, precise temperature control is achieved using a heater and temperature feedback.

In semiconductor cleaning applications a linear nozzle, typically consisting of a plenum and a linear array of orifices is used ([Figure 17.6](#)). When uniform, single pass cleaning is required, the nozzle's width may be set to equal the width of the substrate. The dimensions and spacing of the orifices are selected so as to achieve maximum cleaning efficiency, with minimum cleaning time and minimum gas consumption. For example, in one design 0.4 mm diameter orifices were spaced on a 3.2 mm pitch. The nozzle was directed at an inclined angle (45°) toward the contaminated surface, and was located about 13 mm away from the surface. This distance has been found to provide sufficient time for aerosol formation during the expansion process, as well as for merging of the individual jets prior to their impinging on the surface.

The nozzle plenum conditions are typically 5 bar and ~ 105 K. During the expansion the mixture pressure is dropped into the approximate range 0.07–0.69 bar, and the free jet temperature falls below the corresponding argon freezing point, ≤ 84 K ([Figure 17.4](#)). The substrate is, therefore, exposed to an aerosol jet at cryogenic temperatures under a partial vacuum. A commonly used expanded gas pressure is ~ 0.07 bar. This vacuum creates a slightly

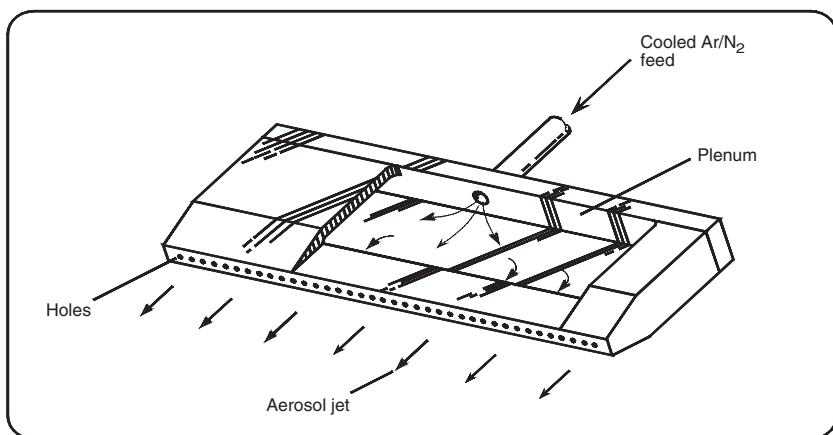


FIGURE 17.6 Cut-away view of linear nozzle.

rarefied atmosphere, which promotes slippage of argon particles through the boundary layer and to the surface. For example, the slip correction factor²⁵ for 1 μm spherical particles at 0.33 bar is $\sim 43\%$ higher than at atmospheric pressure. This results in a 30% reduction in the decelerating drag force by the gas on the particle. This increased particle slip enhances the collision process by allowing the aerosol particle to strike contaminants at a higher velocity, and, thereby, tends to increase the cleaning efficacy. Experimental tests have, in fact, confirmed better removal of silicon oxide debris particles from silicon oxide surfaces under partial vacuum than at atmospheric pressure. However, since the gas phase assists in the removal and venting of released contaminants, expanded gas pressures lower than ~ 0.05 bar are typically not used. A shutter-type vacuum valve can be placed in the vent line, and adjusted to set the pressure of the upstream cleaning chamber.

These operating conditions require the process to be performed in a well-sealed vacuum chamber (Figure 17.3). The chamber effluent stream is re-warmed to near ambient temperature using an electrical resistance heater, immediately after exiting the cleaning chamber. Solid argon at pressures below the triple point pressure is then returned to the gaseous state through sublimation. An oil-free vacuum pump is used with a small nitrogen bleed flow to prevent back streaming of contamination into the cleaning chamber.

The nozzle is held stationary while the substrate is moved on a carrier. A purge gas stream consisting of high purity filtered nitrogen is typically used to create a steady flow from the substrate loading area to the aerosol jet (Figure 17.3). This stream inhibits the movement of released contaminants into the substrate loading area. The substrate, therefore, remains uncontaminated after emerging from the aerosol jet. The ambient temperature purge stream also assists in re-warming the substrate to ambient temperature prior to unloading from the cleaning chamber. Substrate re-warming can also be accomplished using a heated carrier, or a radiant heating device located in the loading area.

The sealed chamber and moving mechanism permit exposure of the substrate to the aerosol jet in a controlled and uniform manner. The sealed chamber also prevents exposure of the cooled substrate to condensable atmospheric substances. Any condensation on the cooled substrate would re-contaminate its surface. The internal geometry of the chamber is designed to produce a smooth flow of cryogenic aerosol from the nozzle, across the substrate surface, and to the vent, without substantial mixing or re-circulation. Such carefully controlled flow patterns cause the released contaminants to travel out the vent opening without progressive accumulation in the chamber. This, in turn, prevents significant re-deposition of contaminants on the substrate surface.

In one commercially available system the wafer is heated by proximity to a heated chuck during aerosol cleaning. Heating of the wafer sets up thermophoretic forces, which minimize re-deposition of particles once they are released from the surface. In addition, the chamber walls are heated to minimize re-deposition of released particles within the cleaning chamber.

It has been found that thermal insulation of an aluminum or stainless steel cleaning chamber is not necessary to maintain aerosol formation. Sufficiently low temperatures occur during gas expansion to form solid argon locally in the free jet. Viewports are usually installed in the cleaning chamber to verify aerosol formation during operation.

Process monitoring consists primarily of pressure, flow rate and temperature measurements at various system locations. Table 17.1 lists the minimal quantities that should be monitored and controlled during operation of a typical cleaning unit.

17.3.2 Operating Conditions

A list of the operating conditions for one example of a 200 mm silicon wafer cleaning system is provided in Table 17.2. The list of utilities required to operate the same system is shown in Table 17.3. Required aerosol flow rates and utilities for this system scale directly with the substrate size. For example, if the nozzle width is set equal to the substrate diameter, the required gas flow rate increases directly with the substrate diameter. When cleaning a 200 mm silicon wafer in the example system, the aerosol is operated for ~48 s during

TABLE 17.1 Process Parameters Monitored	
Measured Quantity	Location
Flow rate	Argon feed gas, nitrogen feed gas, nitrogen purge gas
Temperature	Chamber walls, wafer chuck, liquid nitrogen coolant, vent gas after re-warming
Pressure	Feed gas (nozzle plenum), cleaning chamber, liquid nitrogen coolant

TABLE 17.2 Cleaner Operating Conditions	
Wafers cleaned per hour	50
Cleaning chamber pressure	0.33 bar
Nozzle plenum pressure	5 bar
Nozzle plenum temperature	105 K
Argon flow rate in aerosol	450 l per min
Nitrogen flow rate in aerosol	45 l per min

TABLE 17.3 Cleaner Utilities Requirements

Argon used per wafer cleaning	360 Standard liters at 5 bar
Nitrogen used per wafer cleaning	36 Standard liters at 5 bar
Liquid nitrogen used per wafer cleaning	0.51
Vent	500 Standard liters per minute argon and nitrogen at 0.33 bar
Power	120 V, 2 phase, 100 A
Cleaning tool footprint	1.1 m × 1.6 m (1.7 m high)

the cleaning cycle. During this time a cooling capacity of ~ 1300 W is required in the pre-cooler. The wafer is passed under the aerosol nozzle once in each direction during the cycle. This aerosol exposure time corresponds to wafer travel velocity of about 8 mm per second in each direction.

The above operating conditions are illustrative and can be substantially varied to produce desired results. The cleaning intensity can be controlled through adjustments to the feed gas mixture ratio or temperature, the nozzle expansion ratio, and the duration of surface exposure to the aerosol. For example, during experimental tests, the cleaning chamber pressure has been varied in the range 0.05–1 bar, and the argon/nitrogen mixture ratio has been varied in the range 10:1–1:10. More recently, processes have been developed which use pure nitrogen to form the aerosol.²⁶ Other changes can be made to the nozzle design, orientation and position to produce desired effects.

All the utilities required for the example cleaning system, including high purity argon and nitrogen at 5 bar pressure are normally available in modern semiconductor processing facilities. Since only inert atmospheric gases are used in the process, no vent gas scrubbing or other gas abatement is required. However, the released surface contaminants, normally present in particulate form, must be separated from the inert exhaust gas prior to venting into the atmosphere. This separation can be performed using standard emission control procedures, such as filtration.

17.3.3 Cleaning Systems

Various experimental and commercial argon aerosol systems have been developed for silicon wafer cleaning. These systems have been configured as manually loaded units, cassette-to-cassette tools or cluster tool modules. Each

automated system can process about 40–60 wafers per hour. Wafers up to 300 mm in diameter can be processed by the systems. Although liquid nitrogen is typically used for pre-cooling the feed gas stream, closed-cycle helium refrigerators have also been used commercially. Such refrigerators eliminate the need for a continuous source of liquid nitrogen during operation. Depending upon the system design, operating cycle and feed mixture recipe, the various systems consume about 50–360 standard liters of argon per wafer cleaning.

One manually loaded experimental design is shown in [Figure 17.7](#). The wafer carrier in this unit is moved by a rack and pinion actuator, which is driven by a computer-controlled stepping motor. This system includes a separate load-lock chamber containing a 2000 W rapid thermal processing (RTP) radiant heater. A gate valve separates the cleaning chamber and the load-lock chamber. This system can be converted to an automatically loaded cluster-type tool. In this case, the load-lock chamber isolates the cleaning chamber from the tool's central wafer handling platform, which may be held at a substantially lower operating pressure.

The top view of one cassette-to-cassette system using helium refrigeration is shown in [Figure 17.8](#). The robotic wafer handler places a contaminated wafer onto a carrier located in a buffer chamber. The wafer is then passed under the aerosol nozzle located in the cleaning chamber. The robot returns the cleaned wafer to a cassette chamber. All chambers in the system are vented free of atmospheric contaminants and contain a clean, inert atmosphere. The system shown in [Figure 17.8](#) uses a closed-cycle cryogenic refrigerator for pre-cooling the feed gas. The refrigeration power requirements in such a system increase as the feed gas flow rate increases. In order to minimize the required gas flow rate, fewer orifices are placed along the linear nozzle, and the orifices are spaced farther apart. A raster-type motion is then used in which the silicon wafer is moved

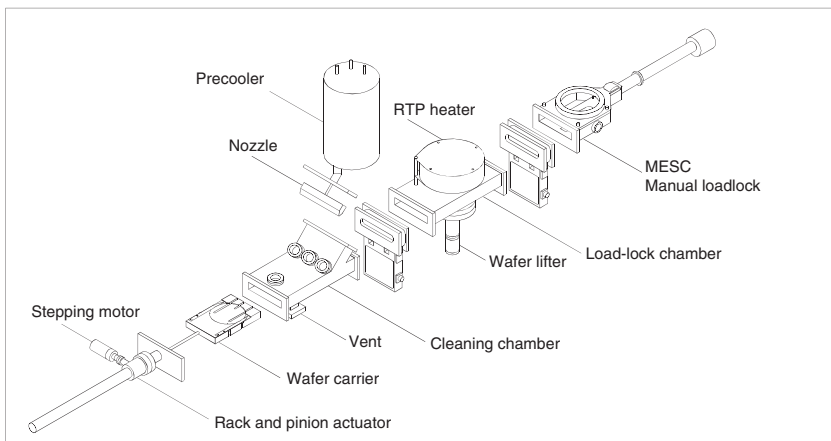


FIGURE 17.7 Exploded view of manually loaded wafer cleaning system.

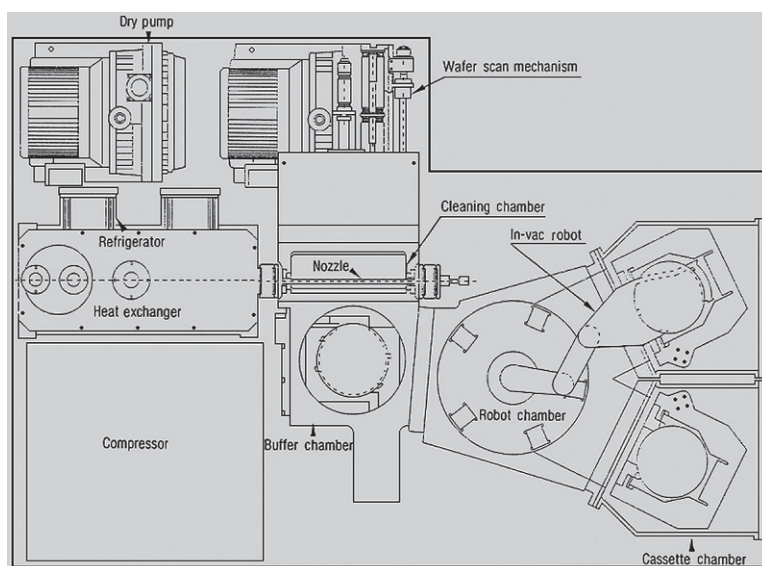


FIGURE 17.8 Top view of cassette-to-cassette wafer cleaning system (Courtesy of Sumitomo Heavy Industries Ltd.)

in a zigzag pattern, rather than a linear motion under the nozzle. In this way the entire wafer surface is exposed to the aerosol during cleaning.

One commercialized system^c includes a novel method to increase the aerosol energy available for contaminant removal. Using a dual-nozzle process (see Figure 17.9), the argon aerosol particles are accelerated to near-sonic velocities before striking the wafer surface. The particles are accelerated by merging the aerosol jet with a second, higher velocity gaseous nitrogen jet. Pre-cooling of the separate nitrogen stream has been found to be unnecessary. Sufficient pre-cooling can be accomplished during the expansion of the accelerator gas to the lower pressure of the cleaning chamber. A converging/diverging nozzle design can be used to achieve supersonic velocities in the accelerator jet. In this system the accelerator nozzle is placed near the aerosol nozzle in the cleaning chamber. The argon aerosol is then entrained in the higher velocity nitrogen jet before striking the substrate. The kinetic energy of an aerosol particle increases with the second power of the particle's velocity. Aerosol acceleration, therefore, provides a means to substantially increase the energy transferred to a surface particle during collision. Test results discussed below suggest that more strongly adhering objects, including fingerprints, photoresist layers, aluminum films, and chemical mechanical planarization (CMP) slurry residues can be removed by the dual-nozzle process. Accelerated aerosols can, therefore, increase the number of possible applications for cryogenic cleaning.

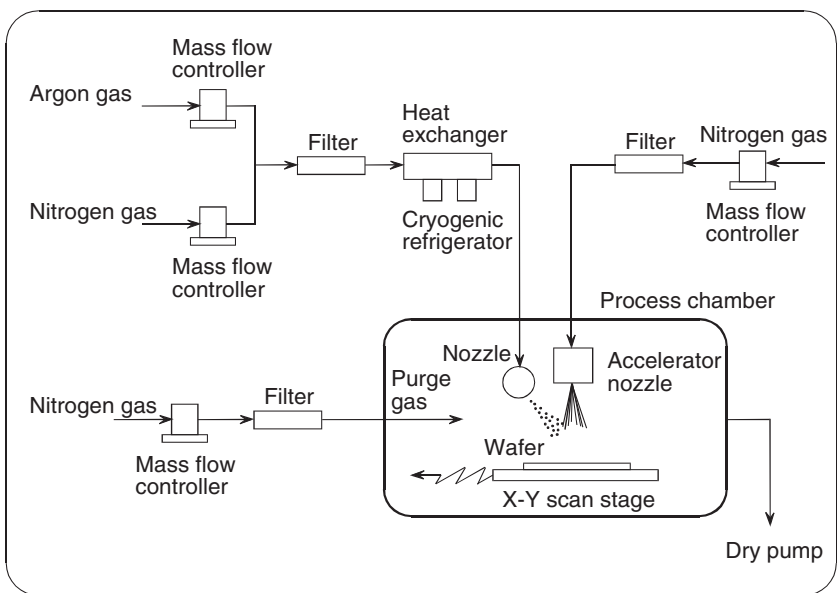


FIGURE 17.9 Accelerated argon aerosol jet cleaning system (Courtesy of Sumitomo Heavy Industries Ltd.)

Another commercially available system is shown in [Figure 17.10](#). While the process chamber and support module for this system are still designed with clustering in mind, the robotics is designed for stand-alone operation. The robotics is designed to accommodate two process chambers to increase overall throughput, but not to perform sequential processing. Since the tool is operated in a stand-alone, single cleaning process mode, vacuum load-locks and vacuum robotics are not necessary. The process chamber is vented to atmospheric pressure to load and unload wafers under an inert nitrogen purge. This significantly reduces the cost of the system and provides for faster wafer loading. Another significant difference from other commercially available systems includes a nozzle length covering the entire diameter of the wafer. This requires only a linear scan of the wafer to clean the entire surface. However, a higher gas flow is needed to create aerosol jet across the entire wafer. This increases the usage rate of liquid nitrogen in the pre-cooler.

17.4 COST

When determining the most appropriate method for any cleaning application, the process cost must be weighed against other factors such as cleaning efficiency, throughput, safety, waste disposal considerations, impact on the environment, availability of cleaning agents, space and utilities requirements, and

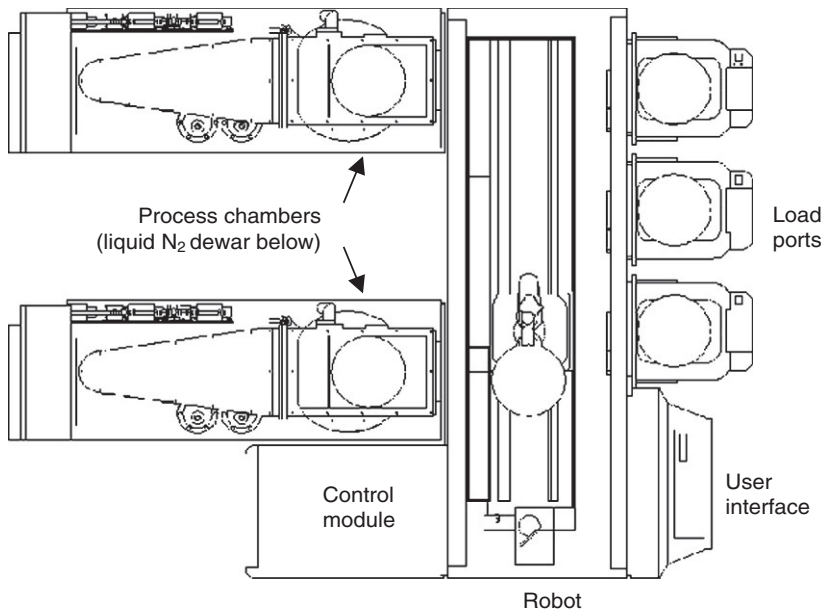


FIGURE 17.10 Top view of cassette-to-cassette wafer cleaning system (Courtesy of FSI International.)

ease of implementation. Argon aerosol surface cleaning requires specialized cryogenic and vacuum hardware, but is cost-effective in semiconductor applications. An approximate cost summary for cryogenically cleaning 200 mm silicon wafers is shown in Table 17.4.²⁷ The total cost per wafer cleaned (US\$0.80) is commensurate with alternative wafer cleaning processes, such as wet etch baths, brush scrubbing, and sonic cleaning. When the costs for liquid chemical disposal are considered, the argon aerosol process becomes even more

TABLE 17.4 Summary of Costs per 200 mm Wafer Cleaning	
Consumables	US \$0.29
Capital	US \$0.23
Facilities	US \$0.12
Labor	US \$0.11
Scrap loss	US \$0.03
Maintenance	US \$0.02
Total	US \$0.80

economically attractive. Further reduction in operating cost may be achieved through process optimization. For example, a shorter exposure time to the aerosol would reduce all feed gas, coolant and electrical utilities requirements. Since consumables represent the largest fraction of the cleaning cost, such improvements may significantly reduce the cost of ownership.

17.5 EFFECTIVENESS AND APPLICATIONS

17.5.1 Kinetics of Cleaning

Precision surface cleaning can be accomplished with aerosol jets when two conditions are met: (1) the threshold for surface particle re-suspension must be reached and (2) the exposure time to the jet must be sufficient to remove the particles. Simple tests may be used to estimate the rate at which surface particles are removed during aerosol jet cleaning. Experiments were performed to determine the fraction of test particles removed from silicon substrates.²⁸ Figure 17.11 shows a typical result. The Tencor 6200 wafer scans show a contaminated silicon wafer before and after exposure to an argon aerosol jet. Before exposure, the 100 mm diameter wafer had ~ 3800 $0.2\ \mu\text{m}$ polystyrene latex spheres distributed across its surface. After cleaning, only 53 particles larger than $0.15\ \mu\text{m}$ were detected by the instrument, corresponding to a removal efficiency of $>98\%$ and a surface density of <1 particle cm^{-2} . This cleaning performance is considered acceptable for some wafer cleaning applications.

The above tests provide information on the kinetics of the aerosol cleaning process. We can, for example, assume that aerosol particles having uniform diameter D collide with surface particles having uniform diameter d . The probability that a given surface particle will encounter a collision with an aerosol particle depends on the cumulative number N_i of incident aerosol particles striking a given surface area, A_s . This number is equal to the product of (1) the flux of aerosol particles striking the surface and (2) the exposure time of the surface to the aerosol:

$$N_i/A_s = (\text{particle flux}) \times (\text{exposure time}) \quad (17.5)$$

If the initial number of contaminant particles on the surface is N_0 , then the number of surface particles, N , remaining after exposure to the aerosol is given by²⁹:

$$N = N_0 \exp(-A_r N_i/A_s) \quad (17.6)$$

where the effective impaction area of each particle on the surface, A_r , is given by $\pi(D+d)^2/4$. Under these assumptions the fraction of remaining surface particles decreases exponentially with particle flux and exposure time to the aerosol. The aerosol particle flux to a surface under typical operating conditions is not known. However, in the above test, the 100 mm silicon wafer was translated under the aerosol jet in <10 seconds. The total exposure time for a given point

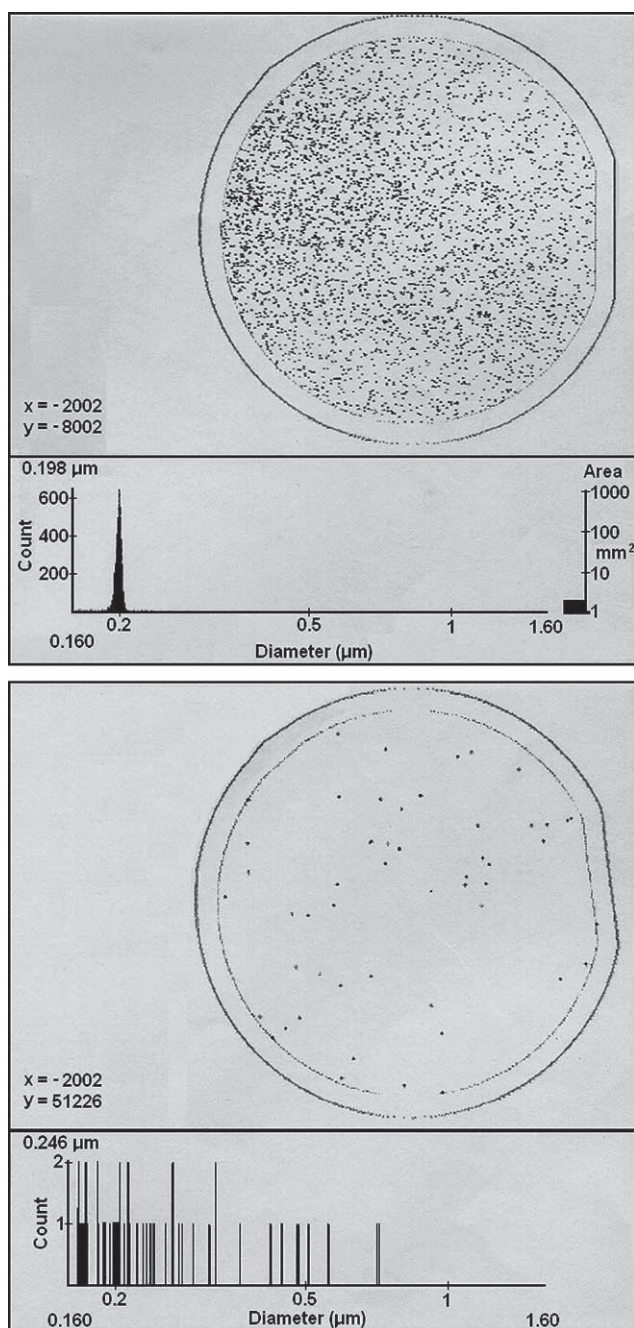


FIGURE 17.11 Tencor 6200 scans of silicon wafer (top) before exposure and (bottom) after exposure to argon aerosol jet.

on the surface to the aerosol was <1 second, and yet more than 98% of the particles were removed ($N/N_0 < 0.02$). If we assume that $D \sim 1 \mu\text{m}$ and $d \sim 0.2 \mu\text{m}$, then we find that the cumulative number of aerosol particles striking the surface, N_i/A , was at least $1.4 \times 10^8 \text{ cm}^{-2}$ during a 1-s exposure.

These tests demonstrated that the energy threshold for removal of the polystyrene latex surface particles was achieved by the colliding argon particles. The tests also showed that the cumulative number of aerosol particles striking the surface was sufficient to provide a high likelihood of collision with sub-micrometer surface particles.

The above surface particles were spherical, quite uniform in size and placed on a flat test surface. Therefore, the variability in adhesion energy for this population of particles was relatively small. Such test particles may not closely represent some ambient or process-generated contaminants. However, in other experiments the composition and geometry of the test particles were varied. In addition to polystyrene latex and glass microspheres, silicon nitride and silicon dioxide particles having irregular shapes were used. The test particles, ranging in size from 0.15 to 20 μm , were deposited on silicon, silicon dioxide, silicon nitride, and photoresist substrates. Similar rapid removal rates were observed in each case, and post-clean particle densities on the surfaces were near 1 cm^{-2} . Ambient or process-generated particles can also be removed rapidly when the threshold for particle re-suspension is achieved. When initially clean wafers were exposed to the argon aerosol jet, no more than five new particles were added to a 100 mm wafer. The new particles may have resulted from trace condensable impurities or particulate contamination in the cleaning chamber. Such new “adders” can be reduced using improved process control and chamber design.

When the threshold for re-suspension of particles [Eq. (17.3)] is met, the removal rate of the particles is determined by kinetic considerations. For example, in one study polystyrene latex spheres as small as 41 nm were used.³⁰ Some decrease in removal efficiency, $1 - N/N_0$, is seen for these very small particle sizes. In another study, silicon nitride and tungsten particles were used at sizes as small as 65 nm.²⁴ These results are shown in Figures 17.12 and 17.13. Tungsten particles were more difficult to remove than silicon nitride particles. In comparison to a wet process using standard cleaning chemicals, it was found that the cryogenic aerosol process had slightly lower removal efficiency for the smallest silicon nitride particles, but higher removal efficiency for the smallest tungsten particles. Although smaller particles have generally lower adhesion energies, as explained in Section 17.2.2, they also present lower effective impaction areas, A_r . The kinetic model represented by Eq. (17.6) shows that such small surface targets can remain in greater numbers after exposure to an aerosol jet. Complete removal of smaller particles, therefore, requires greater exposure time, aerosol flux, or aerosol particle size, D .

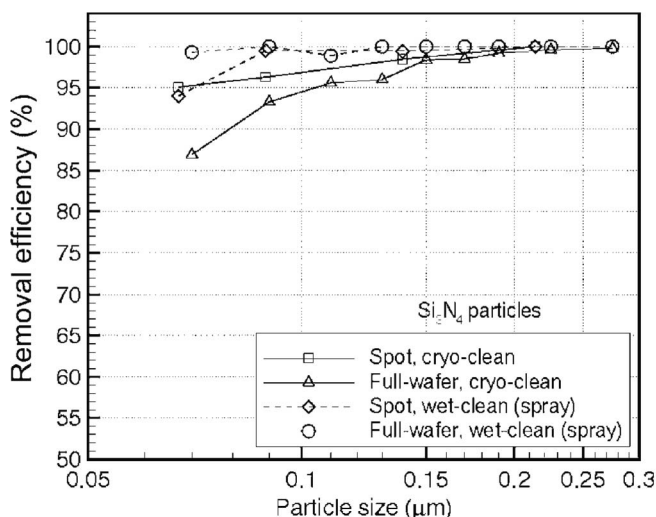


FIGURE 17.12 Removal efficiency of silicon nitride particles as a function of particle size for cryogenic aerosol and a standard wet spray cleaning processes. The terms “spot” and “full wafer” refer to the distribution of the particles over the wafer surface.²⁴

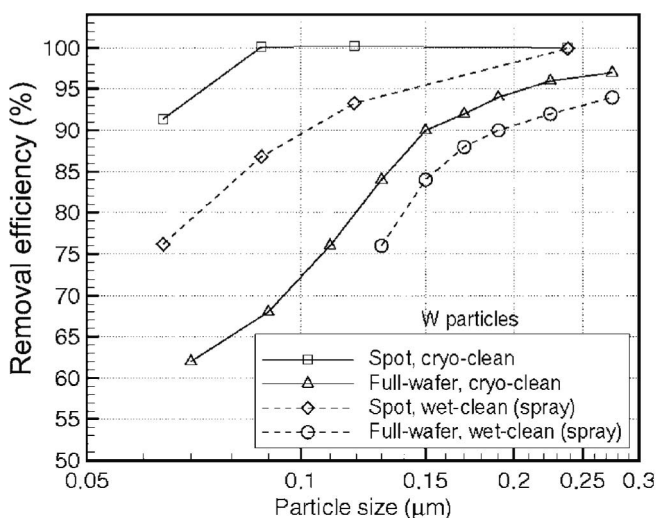


FIGURE 17.13 Removal efficiency of tungsten particles as a function of particle size for cryogenic and a standard wet spray cleaning processes. The terms “spot” and “full wafer” refer to the distribution of the particles over the wafer surface.²⁴

17.5.2 Cleaning Performance

Figure 17.14 shows an SEM view of etched aluminum lines on a microcircuit following removal of the photoresist layer. The lines shown in Figure 17.14 (top) contain sidewall residue and surface contamination (fences). Substantial chlorine byproduct residue resulting from the etching process was also present. Such contaminants would significantly reduce the reliability of the completed

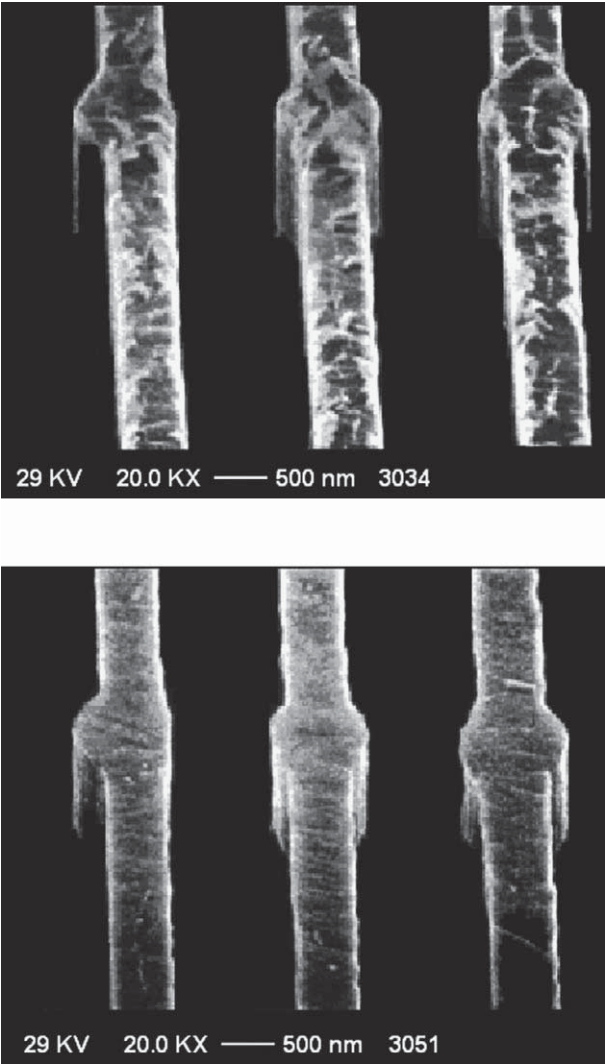


FIGURE 17.14 SEM micrographs showing aluminum lines following reactive ion etching (top) before exposure and (bottom) after exposure to argon aerosol jet.

microcircuit. After cleaning by the argon aerosol (Figure 17.14, bottom figure) the polymer residue was removed, and no chlorine was detected by ion chromatography. Post-etch cleaning of silicon wafers without chemicals using the cryogenic argon/nitrogen aerosol was investigated in the manufacturing environment. It was found, however, that advanced etching and ashing processes left highly oxidized residues that were chemically bound to the surface of the aluminum lines or dielectric surface (in the case of via etching). These tightly bound residues are not removable with an argon/nitrogen cryogenic aerosol, as they require some level of chemical etching. The cryogenic aerosol process has been used in manufacturing, in addition to the chemical cleaning of etch residues, to remove surface debris and significantly improve device yield.³¹ Figure 17.15 shows the types of defects that were removed after the patterning process. Figure 17.16(a-d) is the defect map showing the sum of defects for 36 wafers as they go through the via patterning process. The cryogenic aerosol is able to remove the accumulated debris.

Tiny oil droplets produced by fingerprints are more difficult to remove by argon aerosols. Because of their tendency to spread and, thus, relatively high contact areas, the adhesion energy of such flattened droplets exceeds that of many process-generated particles. Such oily contaminants can be removed using solvents, including liquid carbon dioxide. Alternatively, the aerosol accelerator design described above can be used to increase the available cleaning energy of the aerosol jet. Figure 17.17 shows a patterned silicon wafer with fingerprints, before (top) and after (bottom) exposure to the accelerated aerosol jet. This test shows that the accelerator can be used to tune the jet energy to a level sufficient to remove tightly adhering contaminants, without causing damage to delicate microcircuits.

The argon aerosol has also been found to remove some trace molecular contaminants from a silicon surface. Table 17.5 shows total reflection X-ray fluorescence (TXRF) measurements on a typical silicon wafer surface before

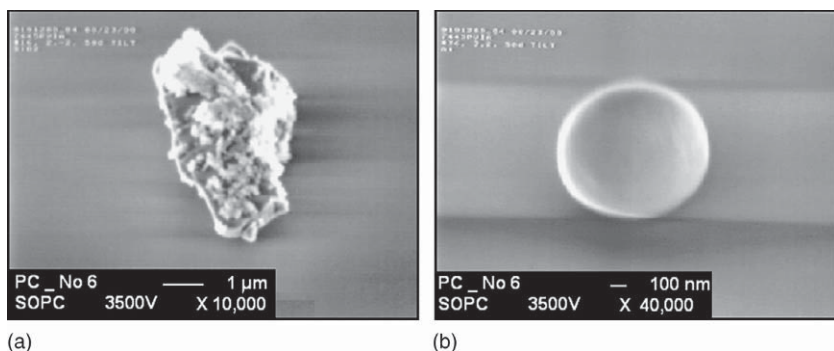


FIGURE 17.15 SEM micrographs of typical defects remaining on the wafer surface after via patterning. The defects were composed of (a) silicon dioxide or (b) aluminum particles.³¹

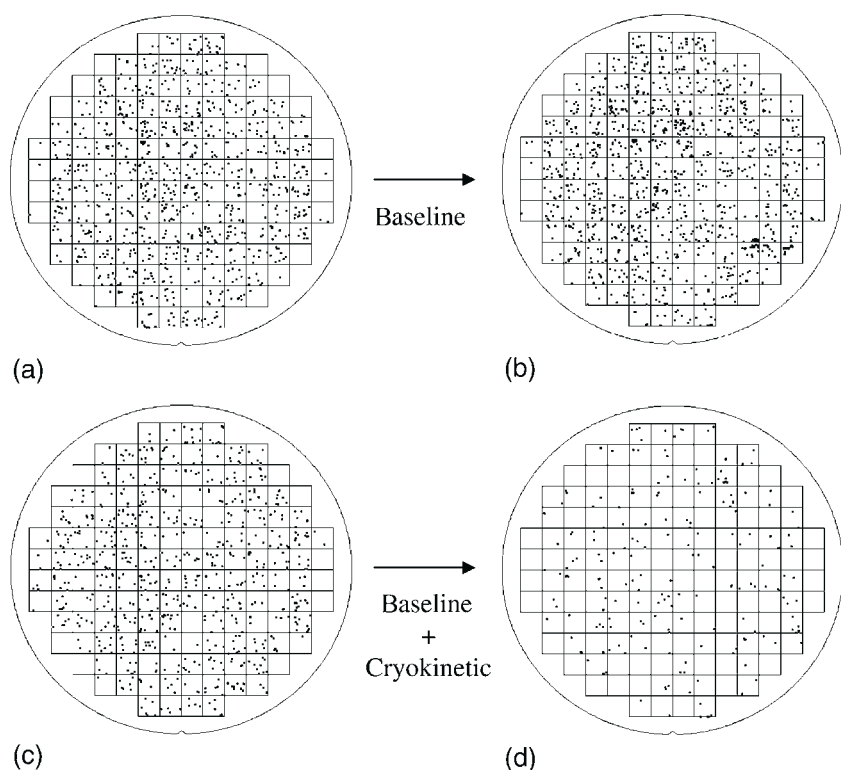


FIGURE 17.16 Defect maps showing the sum of defects from 36 wafers out of three split lots. Map (a) shows the baseline process performance before third-level via etch (average of 26 defects/wafer), and map (b) shows the baseline process performance after via etch (average of 29 defects). Map (c) shows the baseline process with aerosol cleaning before third-level via etch (average of 25 defects), and map (d) shows the baseline process with aerosol cleaning after via etch (average of 7.8 defects).³⁷

exposure to a Standard Clean-1 (SC-1) (water/ammonium hydroxide/hydrogen peroxide) solution. After exposure to the SC-1 solution the same wafer had significantly increased surface concentrations of sulfur, chlorine, calcium, chromium, iron, cobalt, nickel, copper, and zinc. These residues can adversely affect subsequent processing steps and device performance. After exposure to an argon aerosol jet, the concentrations of calcium, chromium, iron, cobalt, and nickel were returned to near their initial levels, and the concentration of chlorine was reduced. Sulfur, copper, and zinc were affected less by the aerosol. The most affected contaminants may have been deposited on the wafer as particles or precipitates, which may be more efficiently removed by the aerosol. Post-SC-1 residue reduction is, therefore, another possible application for argon aerosol surface cleaning.

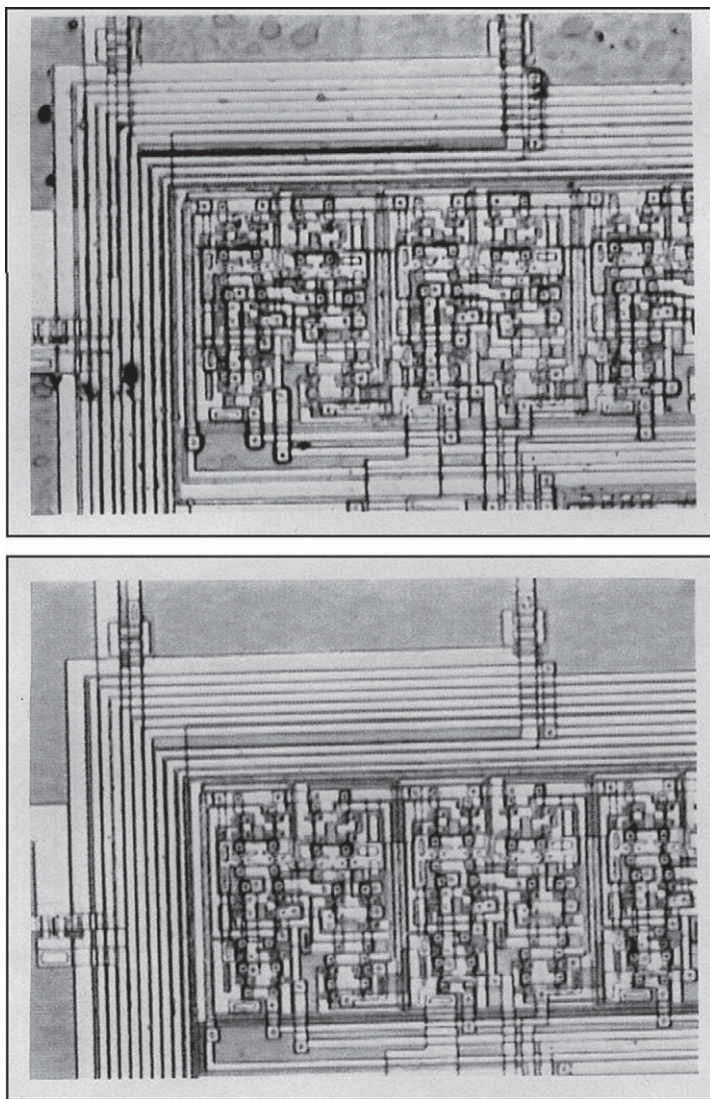


FIGURE 17.17 Light micrographs showing patterned silicon wafer contaminated with fingerprints (a) before exposure and (b) after exposure to accelerated argon aerosol jet.

Particulate contaminants can be removed from within etched surface features using the argon aerosol jet. For example, submicrometer silicon oxide debris particles were removed from inside $1\text{ }\mu\text{m}$ wide by $6\text{ }\mu\text{m}$ deep oxide trenches using the argon aerosol; and $0.05\text{ }\mu\text{m}$ polystyrene latex spheres were removed from within holes having a diameter of $0.25\text{ }\mu\text{m}$ and an aspect ratio of five.³² The physical mechanism of removal in these cases is not understood.

TABLE 17.5 TXRF Data from Silicon Wafer Surface				
Element	Before SC-1 Dip (atoms/cm ² × 10 ¹⁰)	After SC-1 Dip (atoms/cm ² × 10 ¹⁰)	After Aerosol Cleaning (atoms/cm ² × 10 ¹⁰)	Removed (%)
S	5.76	314	293	7
Cl	5.75	68.2	33.3	51
K	4.37	6.65	6.22	6
Ca	0	2.73	0	100
Ti	0	0	0	—
Cr	0	5.69	0.79	86
Fe	0.17	231	1.42	99
Co	0	32.8	0	100
Ni	0.08	1.05	0	100
Cu	2.28	312	284	9
Zn	1.60	24.1	21.1	12

It has been postulated that argon particles approaching at an oblique angle may have sufficient kinetic energy to rebound into microscopic trenches and holes, as shown in [Figure 17.18](#). Removal of surface particles may then result from collision, or by rapid sublimation of a trapped argon particle within the trench, followed by a vapor burst.

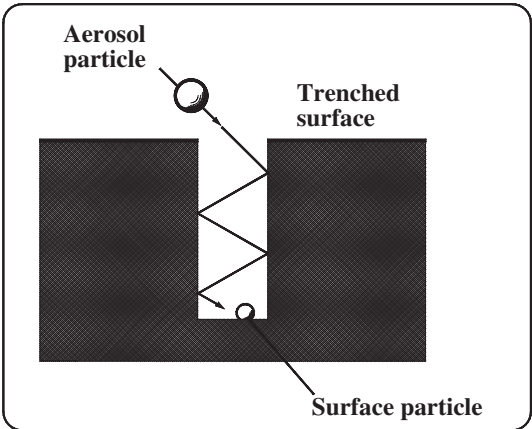


FIGURE 17.18 Possible mechanism for cleaning etched surface features.

Silicon wafers have a high level of surface contamination following chemical mechanical planarization (CMP). The contamination includes significant amounts of slurry material containing submicrometer alumina or silica abrasive particles. Residual materials from the planarized substrate are also present in significant quantities. Such contamination must be removed before further processing can be performed on the wafer. Post-CMP cleaning is typically performed using wet chemical processing. The argon aerosol process has also been found to remove post-CMP contamination from silicon wafers. However, the jet acceleration process is necessary to achieve high removal efficiencies in post-CMP cleaning. In separate tests, both 0.05 μm silica and 0.2 μm alumina slurry particles have been removed from silicon wafers using the accelerated aerosol process. Test results for alumina particles are shown in Figure 17.19. Nearly 100% removal of the slurry particles was achieved when the nitrogen accelerator jet was used. The removal efficiency was also greater at a lower chamber pressure. Lower chamber pressures tend to create solid argon that appears hard and sleet-like, rather than fluffy and snow-like. Studies have also shown that lower chamber pressures lead to smaller aerosol clusters and higher aerosol velocity (Figure 17.20²³). Such conditions appear to enhance the energy transfer to adhering post-CMP contaminants. Figure 17.19 also shows that higher nitrogen purge rates from the wafer loading area tend to more efficiently remove released contaminants from the cleaning chamber, thereby preventing re-deposition of released material, and improving the overall cleaning effectiveness.

Transferring wet, slurry-coated wafers into the dry, vacuum cryogenic aerosol chamber is not practical in a manufacturing environment. However, the

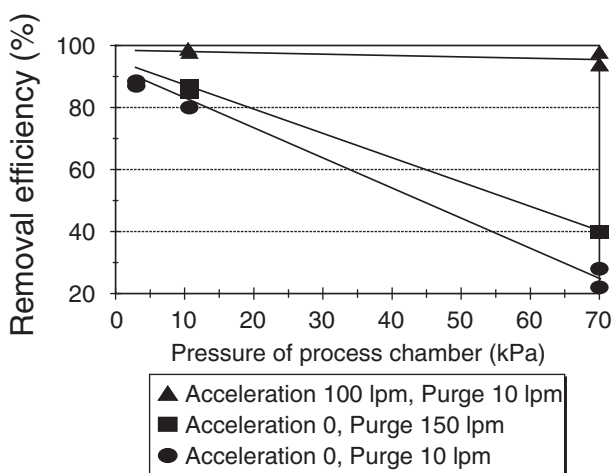


FIGURE 17.19 Accelerated aerosol removal efficiency for wafers contaminated with 0.2 μm Al_2O_3 CMP slurry (lpm = liters per minute) (Courtesy of Sumitomo Heavy Industries Ltd.)

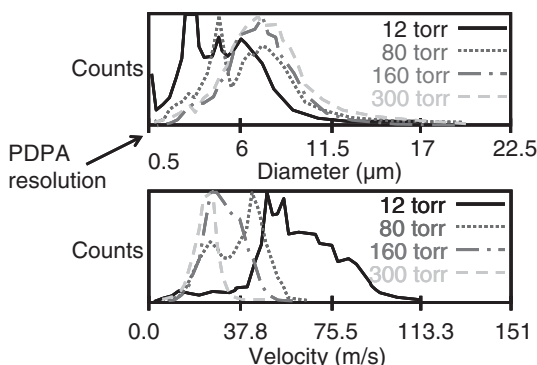


FIGURE 17.20 Size and velocity distributions of a cryogenic aerosol measured by phased Doppler particle analysis (PDPA) for various chamber pressures.

cryogenic aerosol process has found use in final particle removal following the initial post-CMP wet clean operation.³³

Tests have shown that argon aerosol jets do not induce surface damage on bare wafers. No increase in surface roughness, triboelectric charge build-up, or evidence of thermally induced damage, including damage to deposited films, has been detected.³⁴ The aerosol jets did not measurably change the dielectric constants of sensitive low- k films.³⁵ And no change in surface hydrophobicity was observed following cleaning.

While the original processes developed in the mid-1990s were not damaging to 0.25 μm patterns, more advanced devices with narrower structures were found to be sensitive to the aerosol process. Patterned polysilicon gate structures with etched linewidths of <100 nm, as well as low- k trench sidewalls with widths of <200 nm, are damaged by aggressive aerosol processes (Figure 17.21). However, recent process development has resulted in an improved, pure nitrogen aerosol process that is able to maintain high particle removal efficiency while eliminating damage to these sensitive wafer features.²⁶ It is believed that eliminating the high-energy tail of the aerosol distribution eliminated feature damage. This was accomplished by using only nitrogen and by lowering the extent of liquefaction of the gas stream in the pre-cooler [Figure 17.21(b)].

17.5.3 Applications

The cryogenic aerosol process was developed for precision cleaning of flat surfaces in semiconductor applications, including silicon wafers. Cryogenic cleaning can be used to remove ambient particles or process-generated contamination. Some of these applications have already been mentioned in the previous section. Specific process steps at which argon aerosols have been used include pre- and post-deposition of both metal and dielectrics (both PVD and CVD processes), and as part of the post-etch residue removal process for metal,

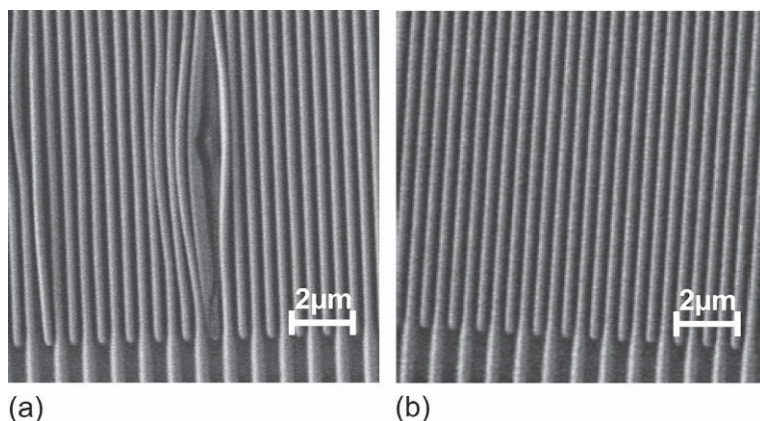


FIGURE 17.21 SEM micrographs of patterned low- k dielectric material showing (a) damage due to high momentum cryogenic aerosol processes and (b) elimination of damage with an optimized cryogenic aerosol process.

polysilicon and dielectric layers. For example, the veil-like residue remaining after photoresist ashing can be removed by argon aerosol.³⁶ The aerosol has also removed partially imbedded particles that remain after sputter deposition. However, since argon aerosol particles have relatively low kinetic energies, the most effective cleaning applications include removal of loosely adhering surface contamination. One such application is at the “back-end” of the microcircuit fabrication sequence. Due to the large number of completed processing steps, microchips have a high value at the back-end. However, electrode probing of the completed microcircuits can cause yield-reducing contamination by metallic particles on the exposed circuits. Conventional wet cleaning processes at this stage can damage the delicate metal interconnects and electrodes of the microcircuits. However, argon aerosol jets have been found to remove the metallic particles without damaging the device. Fine gold lead wires are undamaged by the process.²⁷ Accelerated aerosols, having higher kinetic energies, can also be applied to post-CMP removal of tightly adhering particles and residues.

Cryogenic aerosol has been used in manufacturing for the removal of surface contamination after via patterning for aluminum interconnects with TEOS-deposited silicon oxide dielectric.³¹ The use of a cryogenic aerosol resulted in significant yield improvement (Figure 17.22).

Another application of cryogenic aerosols in manufacturing is prior to dielectric deposition in the copper dual damascene process.^{33,35} Following post-CMP wet processing, polished wafers may still have significant particulate contamination that will affect the integrity of the next dielectric film as well as subsequent processes. Using cryogenic aerosol prior to dielectric deposition removes these yield-limiting particles.

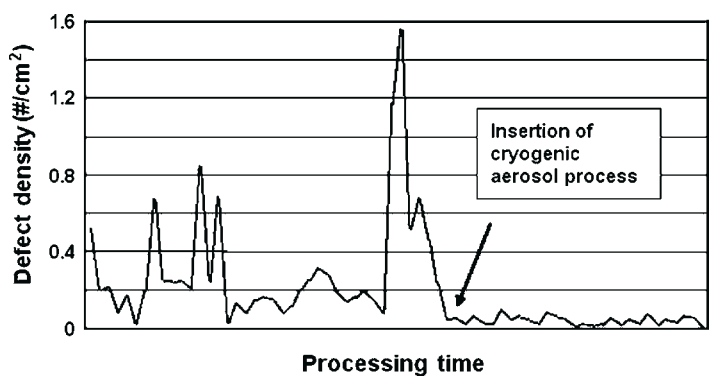


FIGURE 17.22 Trend chart showing significant reduction of in-line TiN barrier defect density ($\#/cm^2$) at the second via level after insertion of the cryogenic aerosol process into the production line.³¹

A further application of the cryogenic aerosol in the manufacturing environment is for the recovery of wafers that have been exposed to fragments of broken wafers. Occasionally, during manufacturing operations, a wafer may break in a load-lock, due to stress from a newly deposited film, or in a carrier, due to mis-handling. Often, these wafers are at a point in the manufacturing sequence where wet chemical processes cannot be used to remove the particles. These wafers must be scrapped at great cost to the manufacturer. It has been shown that wafers with 4000 or more defects of this type can be cleaned with 99% efficiency using the cryogenic aerosol as shown in Figure 17.23.³⁷ Fabrication facilities that are using the cryogenic aerosol for standard cleaning operations have also made use of it to recover valuable wafers subjected to broken wafer fragments.

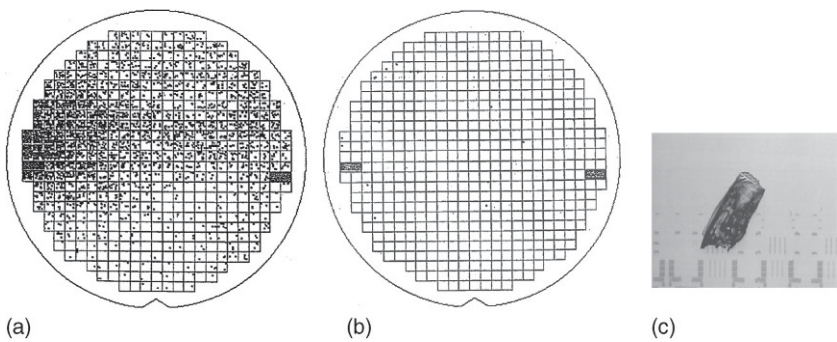


FIGURE 17.23 Defect maps showing removal of broken wafer debris from a metal patterned wafer. (a) The wafer initially had 4284 defects. (b) Cryokinetic cleaning reduced the defect level by over 99% to 13. (c) Typical silicon defect at $10\times$ optical magnification ($\sim 2\ \mu m$ in size).³⁷

With the development of processes with more controlled aerosol momentum,²⁶ new cleaning applications on wafers with patterned features are being implemented in manufacturing.

17.6 FUTURE DIRECTIONS

The argon aerosol jet is one of the newest methods for removing contamination from surfaces. This low temperature process was initially developed as an alternative method to effect environmentally compatible dry surface cleaning of silicon wafers. Well-established applications for the process include particle removal after electrode probing of finished microcircuits, particle removal before line deposition for aluminum interconnect vias, particle removal after gate stack deposition processes, and particle removal after post-metal etch cleaning. More recently discovered applications include cleaning of copper and low- k dielectric surfaces.

Argon/nitrogen cryogenic aerosol surface cleaning can still benefit from further development. The process can be further improved through reduced feed gas and energy consumption, increased cleaning effectiveness and reduced equipment cost. Such improvements may extend the range of applications for this process. Argon aerosol jets may have additional usefulness in flat substrate semiconductor cleaning applications, as well as in non-semiconductor applications. Accelerated aerosols, or other methods designed to increase the kinetic energy or flux of aerosol particles in the jet may be useful in increasing the cleaning efficiency of the argon aerosol. Finally, additional changes to the system hardware intended to simplify the equipment may further reduce cost of ownership.

NOTES

- a. Sumitomo Heavy Industries Ltd., 9–11 Kitashinagawa 5-chome, Shinagawa-ku, Tokyo 141–8686, Japan.
- b. FSI International Inc., 3455 Lyman Boulevard, Chaska, MN 55318, USA.
- c. See note a.

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